Examiner only

8. Sodium is extracted from sodium chloride.

(a) The overall reaction taking place is shown in the equation below.

$$2NaCl \longrightarrow 2Na + Cl_2$$

(i) When carrying out the reaction 120 kg of sodium chloride was found to produce 38.05 kg of sodium.

Calculate the maximum possible mass of sodium that could be produced and use this figure to calculate the percentage yield of this reaction. [4]

$A_r(\text{Na}) = 23$ $A_r(\text{Cl}) = 35.5$

Maximum possible mass = ... kg

Percentage yield = ... %

(ii) Suggest a reason why the yield is less than 100 %. [1]

....

....

(iii) Suggest why this reaction must be carried out under dry conditions. [1]

....

....

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(b) A sample of lithium is found to contain two isotopes.

Isotope	Percentage present in sample (%)
lithium-6	7.59
lithium-7	92.41

(i) Calculate the relative atomic mass (A_r) of lithium. Give your answer to **three** significant figures. [3]

$$A_r = \frac{(\text{isotope 1 mass} \times \text{abundance}) + (\text{isotope 2 mass} \times \text{abundance})}{100}$$

A_r = ...

(ii) State the difference between the atomic structures of lithium-6 and lithium-7. [1]

....

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